

Identifying Undervalued NFL Draft Prospects with Position-Aware Machine Learning

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Thursday, April 30, 2026

1 Introduction

The NFL Draft forces teams to choose among prospects with different athletic profiles, college resumes, and positional roles. The decision is difficult because the relevant outcome is not simply whether a player becomes good in the NFL, but whether he returns more value than was expected from the pick used to acquire him. A productive fifth-round player can be a better draft outcome than a first-round player who performs only slightly above replacement, because the former creates more surplus value relative to cost. This framing is consistent with prior work on draft-market efficiency, which argues that NFL teams often pay too much for the right to make very early selections [3].

Our project asks whether a pre-draft machine learning model can help identify undervalued players. We use draft records and Combine measurements from Pro-Football-Reference for the 2000–2020 NFL Draft classes [6, 5]. Career outcome is measured using weighted Approximate Value (wAV), a Pro-Football-Reference summary measure designed to put player contribution on a common scale across positions and seasons [8]. We then adjust wAV for draft slot to define a surplus-value target.

This connects the project to two strands of prior work. In sports economics, the draft is often studied as a market in which teams trade uncertain future production for scarce draft capital. The value of a pick is therefore not only the expected quality of the player, but the expected quality relative to acquisition cost. In machine learning, the relevant challenge is a noisy tabular prediction problem with a rare positive class. Most drafted players do not become major surplus outcomes, and many important determinants of future success are unobserved. For that reason, the goal is not to make a deterministic scouting oracle. The more realistic goal is to improve ranking and triage: given many prospects with similar public information, the model should help decide which lower-cost players deserve further human attention. This report makes three contributions. First, we define and implement a draft-position-adjusted target, wAVOE, that makes late-round and early-round outcomes comparable. Second, we build a pre-draft modeling pipeline that uses only information available by draft time: round, pick, age, position, weight, and Combine drills. Third, we compare general-purpose models with a position-aware gated neural model that lets the effect of the same physical trait vary by position. The central claim is not that the model should replace scouting, but that it can serve as a decision-support tool for finding small pockets of value where conventional draft expectations are too pessimistic.

2 Methods

2.1 Data and target construction

The cleaned data set contains 5350 drafted players from the 2000–2020 NFL Draft classes. The response variable is based on career wAV. Let d_i be the overall draft pick for player i , and let wAV_i be his observed career weighted Approximate Value. We define

$$\text{wAVOE}_i = \text{wAV}_i - \widehat{E}\{\text{wAV} \mid d_i\}. \quad (1)$$

Here $\widehat{E}\{\text{wAV} \mid d_i\}$ is the historical expected career value for players selected at the same or similar draft position. Positive wAVOE means that a player outperformed the expectation attached to his draft slot; negative wAVOE means that he underperformed that expectation. This is the key modeling choice, because it turns the project from “predict the best players” into “predict players whose value exceeds their acquisition cost.”

We use two prediction tasks. The first is regression on continuous wAVOE. The second is a rare-event classification task for late-round steals. A player is labeled as a steal if

$$\text{wAVOE}_i > 5 \quad \text{and} \quad d_i \geq 100. \quad (2)$$

This label focuses evaluation on the practical draft-room question: among lower-cost players, which prospects are most likely to exceed expectations by a meaningful amount? We deliberately do not use post-draft statistics as input features. Career games, starts, awards, and NFL production are outcomes, not information available to a team on draft day. Excluding them avoids information leakage and keeps the experiment aligned with the actual decision setting.

2.2 Preprocessing and feature engineering

The feature set is restricted to pre-draft information. We include draft round, overall pick, log pick, age, position, weight, and Combine measurements: 40-yard dash, vertical jump, broad jump, three-cone drill, and short shuttle. Since athletic measurements have different meanings across positions, we create within-position z-scores for each Combine metric. Therefore, they represent how unusual a measurement is relative to the player’s own position group.

The data is split into training, validation, and tests sets in a 70/15/15 ratio, with missing Combine values filled by median imputation. This produced 3384 training players, 726 validation players, and 726 test players, stratified by a steal label to maintain a hit rate of approximately 0.136 in each split.

2.3 Models

We compare four models. The first is a random forest, which provides a strong nonlinear baseline and a useful feature-importance check [1]. The random forest is attractive for this problem because it can represent interactions among draft slot, position, and athletic traits without requiring a prespecified linear form.

The second model is a standard multilayer perceptron (MLP). It receives the same engineered numeric and encoded categorical features and has separate output heads for wAVOE regression and steal probability. This model tests whether a flexible neural network improves over the random forest when trained on the same tabular information.

The third model is a position-aware gated neural model. The model embeds each position and uses that embedding to generate feature-wise scale and shift parameters via a feature-wise linear

modulation (FiLM) mechanism [4]. If h is a hidden representation and p is a position embedding, the gated layer computes

$$\tilde{h} = \gamma(p) \odot h + \beta(p), \quad (3)$$

where $\gamma(p)$ and $\beta(p)$ are learned functions of position and $\odot$ denotes elementwise multiplication. In plain language, the model can learn that the same raw trait should be weighted differently for different roles. Speed, agility, and weight need not have the same meaning for quarterbacks, offensive linemen, defensive backs, and receivers. This is the main methodological reason to prefer the gated model over simply adding position as one more one-hot variable. A one-hot position term can shift the baseline prediction for a position, but the gating mechanism can also change the slope-like effect of other features by position. For example, it can learn that a short shuttle time changes the hidden representation more for a defensive back than for a guard, while still training one shared model across the whole data set.

Finally, we evaluate a No-Gate baseline used for an ablation study. This model uses an identical architecture and training protocol to the position-aware gated model but replaces the FiLM mechanism with an identity operation. By removing the position-conditioned scaling and shifting, we can isolate whether the added architectural complexity of gating provides a measurable performance benefit or if the network can learn these interactions implicitly.

2.4 Evaluation

Regression performance is measured with mean absolute error (MAE) on wAVOE. Probability quality is measured with Brier score [2]. Rare-event ranking is measured with precision-recall AUC (PR-AUC), which is more informative than ROC-based summaries when the positive class is rare [7]. This matters because a model that predicts “not a steal” for almost everyone may have acceptable accuracy but still be useless for finding late-round value. We therefore read the metrics together rather than treating any single number as definitive. MAE answers whether the model predicts the surplus-value scale well on average, Brier score answers whether its probabilities are numerically trustworthy, and PR-AUC answers whether it moves true steals toward the top of the review list.

3 Experiments and Results

Table 1: Model performance on the shared evaluation split.

Model	MAE (wAVOE) ↓	Brier score ↓	PR-AUC ↑
Random forest	13.21	0.1469	0.3085
MLP	13.01	0.1123	0.2487
Gated model	13.11	0.2369	0.4167
NoGate Model	13.06	0.2365	0.4184

Table 1 reports the main model comparison. Lower MAE and Brier score are better, while higher PR-AUC is better. The gated model achieves an MAE of 13.11, which is very close to the random forest and MLP. Its Brier score is worse, meaning that its probabilities are less well calibrated. However, it achieves a PR-AUC of 0.4167, second only to the NoGate ablation model and substantially above both baselines.

The results suggest a tradeoff. The MLP has the best Brier score, so its output probabilities are better calibrated. The gated model is less calibrated but better at ranking late-round hits

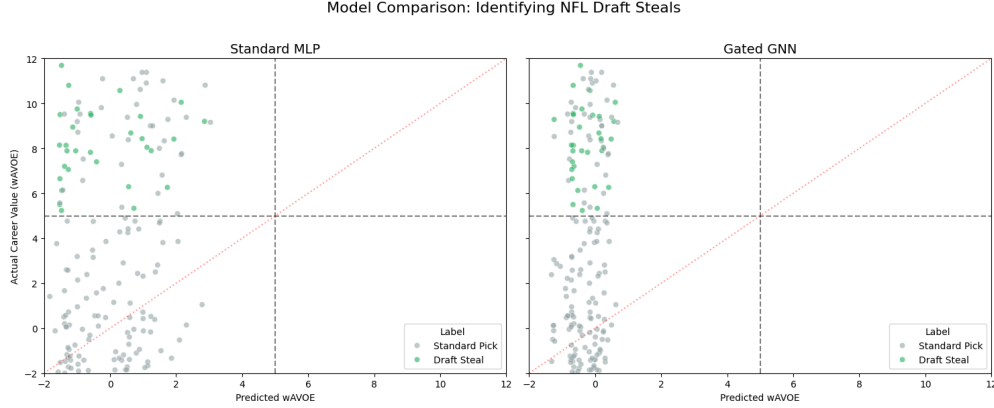


Figure 1: Predicted versus actual wAVOE for late-round players under the MLP and position-aware gated model. This figure visually supports the PR-AUC result by showing how often each model ranks eventual steals above the threshold region.

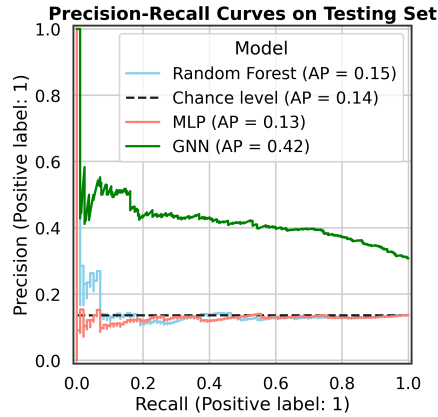


Figure 2: Precision-Recall curves on the test set. The GNN ($AP = 0.42$) substantially outranks the Random Forest ($AP = 0.15$) and MLP ($AP = 0.13$). Test-set AP values differ slightly from Table 1 as those are reported on the validation sets.

near the top of the candidate list. The improvement is substantial in relative terms: compared with the random forest baseline, PR-AUC rises from 0.3085 to 0.4167, and compared with the standard MLP it rises from 0.2487 to 0.4167. The gated model therefore appears to help most on the part of the problem that matters most for a steal detector, even though it is not the best probability-calibration model.

While the gated model achieved a high PR-AUC, the results from our ablation study challenge the importance of complexity when making predictions. The NoGate model performs comparably to the gated model across all metrics, achieving a lower MAE (13.06) and a marginally higher PR-AUC (0.4184). Additionally, it proved more consistent across random seeds, boasting lower standard deviations across metrics. This suggests that while position-awareness is a key theoretical driver, the neural network is processing these interactions implicitly, and the added architectural complexity of the FiLM gate may introduce instability that offsets its benefits.

For a draft-room screening tool, ranking quality is especially valuable. Teams do not need a perfect probability for every prospect; they need a short list of players worth additional scouting attention. In that setting, the gain in PR-AUC is more important than the loss in Brier score,

provided the model is not used as a literal probability forecaster. A natural operational use would be to sort late-round prospects by predicted surplus or predicted steal probability, then send the top group to scouts for film review, injury checks, and scheme-fit discussion.

The exploratory analysis also supports the need for position-aware modeling. Single Combine metrics have only modest marginal correlations with wAVOE. This weak marginal signal explains why simple athletic heuristics are unreliable. At the same time, the correlations are not identical across position groups, and the gated model’s advantage varies by position. The interpretation is that athletic traits are conditionally useful: they matter most when combined with draft cost, role, and other measurements. The model is not discovering one universal rule such as “faster players are always better.” Instead, it learns different trait weightings in different positional contexts.

The model also provides a more actionable view of where value may exist. When analysis is restricted to players selected at pick 100 or later, the model’s most useful predictions are not the average late-round players but the small subset whose predicted surplus is positive and whose observed careers later validated that prediction. Those are the cases that match the practical definition of a draft steal.

4 Limitations

There are several limitations. First, the data are selective. Drafted players and Combine participants are already filtered by scouting processes, so the model is not learning from the full population of college football players. Second, missing Combine data remain a source of uncertainty. Median imputation and missingness indicators are better than deleting incomplete players, but they do not fully recover unobserved athletic information. Third, wAV is only an approximate outcome metric. It is useful because it gives one career-value number across positions, but it cannot fully represent scheme fit, injuries, coaching, roster opportunity, or contract value.

Finally, the results should be viewed as decision support rather than automation. The model can identify players who deserve closer review, but it cannot replace film study, medical evaluation, interviews, or team-specific scheme considerations. A future version of the project should also test temporal generalization more directly, for example by training on older draft classes and evaluating only on later classes. That would better match the real forecasting problem faced by teams.

5 Conclusion

This project reframes NFL Draft prediction around surplus value rather than raw player quality. After adjusting career wAV for draft position, later rounds still contain meaningful positive outliers, which makes undervalued-player detection a realistic machine learning problem. The random forest and MLP provide useful baselines, and both the gated model and its NoGate ablation substantially outperform them on the rare-event detection task.

In plain language, Combine numbers alone are not enough — athletic traits are only meaningful in positional context. The ablation study further shows that much of the gated model’s gain may arise from the neural architecture itself rather than the explicit FiLM mechanism, suggesting positional interactions are partially learned implicitly. The practical contribution is a draft-board screening framework that highlights lower-cost prospects whose profiles deserve closer attention before teams spend scarce draft capital.

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